

Graphical Abstract

Hand Gesture Recognition Using Convolutional Neural Networks for Reliable Human–Computer Interaction

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Highlights

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- Developed a lightweight CNN for automatic hand gesture recognition.
- Used approximately 13,000 RGB gesture images from the Kaggle dataset.
- Achieved approximately 99.15% validation accuracy.
- Evaluated performance using confusion matrix, precision, recall and F1-score.
- Demonstrated reliable human–computer interaction using computer vision.

Hand Gesture Recognition Using Convolutional Neural Networks for Reliable Human–Computer Interaction

Mohammed Aiman Sadhik^a

^a*Department of Business, University of Europe for Applied Sciences, Think Campus,
Konrad-Zuse-Ring 11, Potsdam, 14469, Brandenburg, Germany*

Abstract

Hand gesture recognition has become one of the fastest-growing research areas in computer vision because it enables natural communication between humans and intelligent computer systems without requiring physical contact with conventional input devices. Gesture-based interfaces are increasingly being integrated into robotics, healthcare, gaming, smart homes, industrial automation, virtual reality, and autonomous systems where traditional interaction methods are either inconvenient or inefficient.

Although deep learning has significantly improved image classification performance, many existing gesture recognition systems rely heavily on computationally expensive transfer-learning models requiring substantial processing power and memory. Developing an efficient custom Convolutional Neural Network capable of achieving high classification accuracy while maintaining computational simplicity therefore remains an important research problem.

This study presents a lightweight Convolutional Neural Network framework developed using TensorFlow and Keras for automatic hand gesture recognition. Approximately 13,000 labelled RGB images obtained from the publicly available Kaggle Hand Gesture Recognition Dataset were resized to 128×128 pixels, normalized, and divided into training and validation datasets using an 80:20 ratio. The proposed CNN consists of three convolutional blocks followed by max-pooling, dropout regularization, fully connected layers, and a Softmax output layer.

Experimental performance was evaluated using validation accuracy, precision, recall, F1-score, validation loss, confusion matrix analysis, classification report, and qualitative prediction visualization. The developed CNN achieved a validation accuracy of approximately 99.15%, while precision, re-

call, and F1-score reached approximately 0.99 across all gesture classes. The confusion matrix demonstrated very few classification errors, indicating excellent model generalization.

The proposed framework demonstrates that a carefully designed lightweight CNN can achieve excellent gesture recognition performance without relying on computationally intensive transfer-learning architectures. The developed system provides a practical foundation for future real-time gesture recognition applications supporting reliable and intuitive human–computer interaction.

Keywords:

Hand Gesture Recognition, Convolutional Neural Network, Computer Vision, Deep Learning, TensorFlow, Human–Computer Interaction

1. Introduction

Hand gesture recognition has emerged as one of the fastest-growing research areas in computer vision and artificial intelligence because it enables natural communication between humans and intelligent systems without relying on traditional input devices such as keyboards, computer mice, or touchscreens. Recent advances in deep learning, particularly Convolutional Neural Networks (CNNs), have significantly improved the capability of machines to automatically recognize complex visual patterns from digital images. Consequently, gesture recognition systems have become increasingly important in robotics, healthcare, industrial automation, virtual reality, gaming, smart homes, and intelligent human–computer interaction.

Deep learning has revolutionized image classification by enabling neural networks to automatically learn hierarchical visual features directly from image data. Compared with conventional machine learning methods that require handcrafted feature extraction, CNNs provide higher recognition accuracy, improved robustness, and better generalization across diverse imaging conditions. These advantages have made CNN-based image classification one of the most widely adopted approaches for visual recognition tasks.

Despite these advances, designing an efficient and computationally lightweight gesture recognition system remains an important research challenge. Many existing studies rely on large transfer-learning architectures that achieve excellent classification performance but require considerable computational resources, making them less suitable for real-time deployment on resource-

constrained devices. Therefore, developing lightweight CNN models capable of maintaining high predictive performance while reducing computational complexity has become an active area of research.

This study presents a lightweight Convolutional Neural Network framework for automatic hand gesture recognition using the publicly available Kaggle Hand Gesture Recognition Dataset. The proposed methodology includes image preprocessing, CNN model development, supervised learning, and comprehensive performance evaluation using validation accuracy, precision, recall, F1-score, confusion matrix analysis, and classification reports. The remainder of this section introduces the application background, machine learning context, research gap, objectives, contributions, and overall organization of the study.

1.1. Problem Statement

Automatic hand gesture recognition remains a challenging computer vision problem because gesture appearance varies significantly due to changes in illumination, hand orientation, background complexity, camera viewpoint, and individual user characteristics. These variations often reduce classification accuracy and limit the practical deployment of gesture recognition systems in real-world environments.

Although deep learning has significantly improved image classification performance, many existing gesture recognition systems depend on large transfer learning models that require considerable computational resources. Such models may not be appropriate for applications requiring fast inference and deployment on resource-constrained hardware. Therefore, there is a need to develop a lightweight CNN architecture capable of accurately recognizing hand gestures while maintaining computational efficiency.

The problem addressed in this study is the design and evaluation of a custom Convolutional Neural Network for recognizing five different hand gesture classes using RGB images obtained from the Kaggle Hand Gesture Recognition Dataset. The proposed system aims to achieve high classification accuracy while reducing computational complexity and maintaining suitability for future real-time human-computer interaction applications.

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1.2. Application Background and Practical Importance

Hand gesture recognition has become an increasingly important research area within computer vision because it enables intuitive and touch-free com-

60 munication between humans and intelligent computer systems. Instead of
61 relying on conventional input devices such as keyboards, computer mice, or
62 touchscreens, users can interact naturally through hand movements that are
63 captured using standard RGB cameras. Such interaction methods are par-
64 ticularly valuable in environments where physical contact is inconvenient,
65 unsafe, or impractical.

66 The application domain of this research spans several industries includ-
67 ing healthcare, robotics, industrial automation, virtual reality, augmented
68 reality, gaming, smart homes, automotive systems, and assistive technolo-
69 gies. In healthcare, surgeons and medical staff can control imaging systems
70 without touching contaminated equipment, thereby reducing the risk of in-
71 fection. Gesture recognition is also widely adopted in rehabilitation systems
72 to monitor patient exercises and improve physical therapy. Within industrial
73 environments, workers can operate machinery through gesture commands
74 while wearing protective equipment, reducing reliance on physical control
75 panels. Similarly, gesture-based interfaces improve accessibility for elderly
76 users and individuals with physical disabilities by providing a more natural
77 method of communication with intelligent systems.

78 Traditional human-computer interaction relies heavily on manual input
79 devices such as keyboards, touchscreens, joysticks, and remote controllers.
80 Although these interfaces are effective in many applications, they require
81 physical contact and are often unsuitable in environments where hygiene,
82 mobility, or user accessibility are important considerations. Furthermore,
83 manual interaction may reduce operational efficiency in situations requiring
84 rapid decision making or hands-free control. Wearable sensor-based gesture
85 recognition systems have also been investigated; however, such approaches
86 require additional hardware, increase deployment cost, and often reduce user
87 comfort.

88 Automated image analysis using deep learning provides an attractive al-
89 ternative because modern Convolutional Neural Networks (CNNs) can au-
90 tomatically extract meaningful visual features directly from RGB images.
91 Compared with conventional image-processing techniques that depend on
92 handcrafted feature extraction, CNNs provide higher robustness against vari-
93 ations in illumination, background complexity, hand orientation, and camera
94 viewpoint. The increasing availability of inexpensive cameras together with
95 advances in deep learning has made vision-based gesture recognition both
96 practical and economically feasible for real-world deployment.

97 The rapid growth of intelligent devices further highlights the importance

98 of gesture recognition technologies. According to recent industry reports,
99 billions of smart devices equipped with embedded cameras are currently de-
100 ployed worldwide, creating substantial demand for reliable touch-free interac-
101 tion systems. Consequently, lightweight and computationally efficient CNN
102 models capable of delivering high recognition accuracy have become an im-
103 portant research direction for future intelligent human-computer interaction
104 systems.

105 Figure 1 illustrates several representative application areas where auto-
106 matic hand gesture recognition can improve user interaction and operational
107 efficiency.

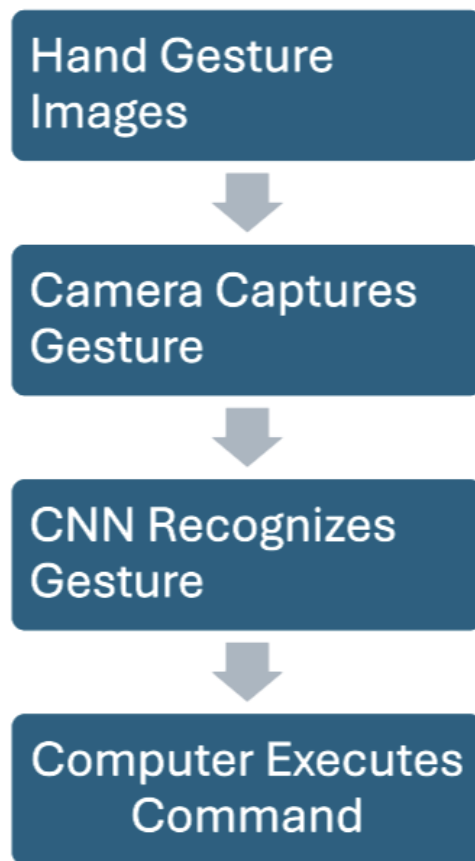


Figure 1: Representative application areas of vision-based hand gesture recognition including healthcare, robotics, smart homes, gaming, and industrial automation.

- 108 • To analyze and preprocess the selected Kaggle Hand Gesture Recognition Dataset.
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- 110 • To develop a lightweight Convolutional Neural Network (CNN) for classifying five hand gesture categories.
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- 112 • To train the proposed CNN using supervised deep learning techniques.
- 113 • To evaluate the model using validation accuracy, precision, recall, F1-score, confusion matrix, and classification report.
- 114
- 115 • To analyze the training behaviour and overall performance of the proposed CNN.
- 116
- 117 • To assess the suitability of the developed model for future real-time human-computer interaction applications.
- 118

119 1.3. Machine Learning and Computer Vision Context

120 Machine learning has transformed computer vision by enabling computers to automatically learn meaningful patterns from digital images without
 121 relying on manually designed feature extraction methods. In image classification tasks, machine learning algorithms analyse pixel-level information
 122 and identify discriminative visual characteristics that distinguish one object category from another. For hand gesture recognition, these characteristics
 123 include the shape of the hand, finger positions, palm orientation, texture, edges, and spatial relationships between different parts of the hand.

124 Among various deep learning techniques, Convolutional Neural Networks (CNNs) have become the most widely adopted architecture for image analysis
 125 because of their ability to automatically extract hierarchical visual features directly from raw images. Unlike traditional machine learning approaches
 126 that depend on handcrafted descriptors such as Histogram of Oriented Gradients (HOG), Scale-Invariant Feature Transform (SIFT), or Local Binary
 127 Patterns (LBP), CNNs learn both low-level and high-level image representations during the training process. Early convolutional layers detect simple
 128 features such as edges and corners, while deeper layers recognise more complex structures including finger arrangements and complete hand gestures.

129 For the proposed hand gesture recognition system, CNNs provide several important advantages. First, automatic feature extraction eliminates the
 130 need for manual image engineering, reducing development complexity and
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141 improving model generalisation. Second, CNNs are capable of recognising
142 gesture patterns despite moderate variations in illumination, background,
143 camera angle, and hand orientation. Third, their hierarchical learning capa-
144 bility enables accurate classification of the five gesture categories used in this
145 study: Backward, Forward, Left, Right, and Unknown.

146 Deep learning is particularly suitable for this project because the Kag-
147 gle Hand Gesture Recognition Dataset contains thousands of labelled RGB
148 images that allow supervised learning of gesture representations. During
149 training, the CNN progressively learns discriminative visual patterns associ-
150 ated with each gesture class and uses this knowledge to classify previously
151 unseen images. This ability to generalise beyond the training dataset makes
152 CNNs highly effective for practical gesture recognition applications.

153 Despite these advantages, deep learning models require sufficient labelled
154 data to achieve high predictive performance. Limited datasets may increase
155 the risk of overfitting, where the model memorises training samples instead
156 of learning general visual characteristics. To minimise this problem, the
157 proposed framework employs image normalisation, dataset shuffling, and an
158 80:20 training-validation split, enabling the CNN to learn robust gesture
159 features while maintaining good generalisation performance on unseen data.

160 1.4. *Transfer Learning for the Selected Problem*

161 Transfer learning has become one of the most widely adopted approaches
162 in deep learning because it enables neural networks to reuse knowledge learned
163 from very large image datasets such as ImageNet. Instead of training every
164 network parameter from random initialization, transfer learning starts with
165 a model that has already learned general visual features from millions of
166 images. These pre-trained models can then be adapted to a new image clas-
167 sification task using considerably fewer training samples and shorter training
168 times.

169 Training a deep Convolutional Neural Network entirely from scratch re-
170 quires a large labelled dataset and significant computational resources. When
171 only a limited number of training images are available, randomly initialized
172 models are more likely to suffer from overfitting, unstable convergence, and
173 lower classification accuracy. Transfer learning addresses these challenges by
174 providing a strong initial set of feature representations that improve learning
175 efficiency and model generalisation.

176 Pre-trained networks learn universal visual features during their initial
177 training. Early layers identify edges, textures, colours, and simple geomet-

178 ric patterns, while deeper layers recognise more complex object structures.
179 During transfer learning, these learned representations are reused for the tar-
180 get application through feature extraction or fine-tuning. Feature extraction
181 keeps most of the pre-trained layers fixed while training only the final classi-
182 fication layers, whereas fine-tuning allows selected layers to be updated using
183 the new dataset.

184 Popular transfer-learning architectures include VGG16, ResNet50, Mo-
185 bileNetV2, DenseNet121, and EfficientNet. These models have demonstrated
186 excellent performance across a wide range of computer vision applications
187 because they combine powerful feature extraction with strong classification
188 capability. However, they generally require more computational resources
189 than lightweight custom CNN architectures.

190 Although the primary focus of this study is the development of a custom
191 Convolutional Neural Network for five-class hand gesture recognition, trans-
192 fer learning provides an important benchmark for comparison. Lightweight
193 CNN models remain attractive because they require fewer parameters, shorter
194 training times, and lower memory consumption while still achieving high
195 recognition accuracy. Consequently, this study investigates whether a care-
196 fully designed custom CNN can provide reliable gesture recognition perfor-
197 mance without depending on computationally expensive transfer-learning ar-
198 chitectures.

199 *1.5. Explainable AI and Trustworthiness*

200 As deep learning models become increasingly complex, understanding
201 how they arrive at their predictions has become an important aspect of
202 trustworthy Artificial Intelligence (AI). Although classification accuracy is
203 commonly used to evaluate model performance, accuracy alone does not
204 guarantee that a model has learned meaningful visual features. A CNN
205 may achieve high accuracy by relying on irrelevant background information,
206 lighting conditions, or other dataset-specific characteristics rather than the
207 actual hand gesture. Such behaviour is known as background bias or short-
208 cut learning and may significantly reduce model reliability when applied to
209 real-world environments.

210 For hand gesture recognition, it is important to ensure that the CNN
211 focuses on the hand region, finger positions, palm orientation, and gesture
212 shape instead of surrounding objects or backgrounds. If the model learns
213 incorrect visual cues, its predictions may become unreliable when images are
214 captured under different lighting conditions, camera angles, or backgrounds.

215 Therefore, evaluating only numerical performance metrics such as accuracy,
216 precision, recall, and F1-score is insufficient for assessing the trustworthiness
217 of a deep learning model.

218 Explainable Artificial Intelligence (XAI) provides methods that improve
219 the transparency of deep learning models by highlighting the image regions
220 that contribute most strongly to a prediction. One widely used technique is
221 Gradient-weighted Class Activation Mapping (Grad-CAM), which generates
222 heatmaps showing the areas of an image that influence the CNN’s decision.
223 These visual explanations allow researchers to verify whether the network
224 is focusing on meaningful hand regions rather than irrelevant background
225 features.

226 Another important explainability technique is SHapley Additive exPana-
227 tions (SHAP). SHAP estimates the contribution of individual image regions
228 toward the final prediction by assigning positive or negative importance val-
229 ues. Unlike Grad-CAM, which primarily provides spatial attention maps,
230 SHAP offers a more detailed interpretation of feature importance and en-
231 ables comparison between correctly and incorrectly classified samples.

232 Although Grad-CAM and SHAP were not implemented as part of the
233 present project, they represent valuable techniques for future model valida-
234 tion and improvement. Incorporating explainable AI into hand gesture recog-
235 nition systems can increase user confidence, improve model transparency,
236 and assist developers in identifying potential biases or weaknesses within the
237 trained CNN. Such interpretability is particularly important for applications
238 involving healthcare, industrial automation, assistive technologies, and other
239 safety-critical human–computer interaction systems where reliable and un-
240 derstandable predictions are essential. ““

241 1.6. Research Gap

242 Although numerous studies have demonstrated the effectiveness of deep
243 learning for hand gesture recognition, several research challenges remain.
244 Most existing systems rely on large transfer learning architectures such as
245 VGG16, ResNet50, EfficientNet, and DenseNet, which achieve excellent clas-
246 sification accuracy but require considerable computational resources and long
247 training times. These models are often unsuitable for deployment on mobile
248 devices, embedded platforms, or other resource-constrained systems.

249 Furthermore, many published studies focus primarily on maximizing clas-
250 sification accuracy while providing limited discussion regarding computa-
251 tional efficiency, model complexity, and deployment feasibility. Relatively

few investigations evaluate whether lightweight custom CNN architectures can achieve comparable performance without depending on pre-trained networks. Another limitation is that many studies report only overall accuracy, with insufficient analysis using precision, recall, F1-score, confusion matrices, and class-wise performance metrics.

The present study addresses these limitations by proposing a lightweight Convolutional Neural Network for five-class hand gesture recognition. The proposed framework emphasizes computational simplicity while maintaining high predictive performance through comprehensive experimental evaluation.

1.7. Research Questions

The following research questions guide the development and evaluation of the proposed hand gesture recognition framework.

- **RQ1:** Can a lightweight Convolutional Neural Network accurately classify five different hand gesture categories using RGB images?
- **RQ2:** How effective are image preprocessing techniques such as resizing and normalization in improving CNN performance?
- **RQ3:** Does the proposed CNN achieve reliable classification performance across all gesture classes?
- **RQ4:** Can the proposed lightweight CNN provide competitive performance without relying on computationally expensive transfer learning architectures?
- **RQ5:** Which evaluation metrics best demonstrate the effectiveness of the proposed hand gesture recognition system?

1.8. Contributions and Novelty

The major contributions of this research are summarized as follows.

- Development of a lightweight Convolutional Neural Network for automatic five-class hand gesture recognition.
- Implementation of a complete image preprocessing pipeline including resizing, normalization, dataset partitioning, and supervised learning using TensorFlow and Keras.

- Comprehensive experimental evaluation using validation accuracy, precision, recall, F1-score, confusion matrix analysis, classification report, and qualitative prediction results.
- Demonstration that a custom CNN architecture can achieve excellent classification performance while maintaining lower computational complexity than many transfer learning approaches.
- Provision of a practical framework that may support future touch-free human–computer interaction systems in healthcare, robotics, industrial automation, smart environments, and assistive technologies.

1.9. Report Organization

The remainder of this paper is organized as follows. Section 2 reviews the existing literature related to hand gesture recognition, deep learning, Convolutional Neural Networks, transfer learning, image preprocessing, and Explainable Artificial Intelligence. Section 3 describes the proposed methodology, including the dataset, preprocessing techniques, CNN architecture, training procedure, and evaluation metrics. Section 4 presents the experimental results obtained from the proposed model together with quantitative and qualitative performance analysis. Section 5 discusses the findings, practical implications, limitations, and future research directions. Finally, Section 6 concludes the study by summarizing the major contributions and highlighting opportunities for future work. ““

2. Literature Review

2.1. Image-Based Machine Learning in the Application Domain

Hand gesture recognition has become an active research area in computer vision because it enables intuitive communication between humans and intelligent systems through visual analysis of hand movements. Image-based gesture recognition systems use digital cameras to capture RGB images and apply machine learning algorithms to identify predefined gesture classes. Compared with wearable sensor-based approaches, vision-based methods are more cost-effective, easier to deploy, and require no additional hardware, making them suitable for applications in healthcare, robotics, smart homes, industrial automation, virtual reality, and assistive technologies.

314 Early hand gesture recognition systems relied on traditional image-processing
315 techniques such as skin colour segmentation, contour extraction, edge detec-
316 tion, and handcrafted feature descriptors. These methods generally per-
317 formed well in controlled environments but often struggled under variations
318 in illumination, background complexity, camera viewpoint, and hand orien-
319 tation. Their performance also depended heavily on manually engineered
320 features, limiting their ability to generalize to new datasets.

321 The introduction of machine learning algorithms such as Support Vector
322 Machines (SVM), Decision Trees, Random Forests, and Artificial Neural Net-
323 works improved gesture recognition performance by learning discriminative
324 patterns from labelled data. However, these techniques still required hand-
325 crafted image descriptors including Histogram of Oriented Gradients (HOG),
326 Local Binary Patterns (LBP), and Scale-Invariant Feature Transform (SIFT).
327 Designing effective handcrafted features required significant domain expertise
328 and often reduced model robustness under real-world conditions.

329 Recent advances in deep learning have transformed image-based gesture
330 recognition by enabling Convolutional Neural Networks (CNNs) to automat-
331 ically learn hierarchical visual representations directly from raw image data.
332 Instead of relying on manually extracted features, CNNs automatically iden-
333 tify edges, textures, shapes, and increasingly complex gesture characteristics
334 during training. This automatic feature learning capability has significantly
335 improved recognition accuracy while reducing dependence on manual feature
336 engineering.

337 Consequently, CNN-based image classification has become the dominant
338 approach for modern hand gesture recognition systems because of its high
339 predictive performance, scalability, and suitability for practical human–computer
340 interaction applications.

341 *2.2. CNN-Based Methods for Hand Gesture Recognition*

342 Convolutional Neural Networks (CNNs) have become the most widely
343 used deep learning architecture for image classification because they auto-
344 matically learn hierarchical visual features from raw image data. Unlike
345 traditional machine learning methods that depend on manually designed im-
346 age descriptors, CNNs perform feature extraction and classification within a
347 single end-to-end learning framework. This capability has significantly im-
348 proved recognition accuracy across a wide range of computer vision applica-
349 tions, including object detection, facial recognition, medical image analysis,
350 and hand gesture recognition.

351 A typical CNN consists of multiple convolutional layers, pooling layers,
352 activation functions, and fully connected layers. Convolutional layers learn
353 local image patterns such as edges, corners, and textures, while pooling lay-
354 ers reduce feature dimensions and improve computational efficiency. As the
355 network becomes deeper, higher-level features representing finger positions,
356 hand orientation, and gesture shape are automatically learned. Finally, fully
357 connected layers combine these learned features to classify the input image
358 into predefined gesture categories.

359 Several researchers have successfully applied CNNs to hand gesture recog-
360 nition. Early CNN models demonstrated that deep learning significantly
361 outperformed traditional classifiers using handcrafted features. More recent
362 studies have introduced deeper architectures capable of achieving excellent
363 classification accuracy even under varying illumination, background com-
364 plexity, and camera viewpoints. These studies confirm that CNNs provide
365 greater robustness and improved generalization compared with conventional
366 machine learning techniques.

367 CNN-based gesture recognition has been successfully applied in numerous
368 practical applications including robotic control, virtual reality, augmented
369 reality, healthcare monitoring, industrial automation, and assistive technolo-
370 gies. The ability to automatically learn discriminative visual representations
371 enables CNNs to recognize complex gesture patterns while reducing the need
372 for manual feature engineering.

373 Despite their strong performance, deep CNN models often require con-
374 siderable computational resources and large training datasets. Consequently,
375 lightweight CNN architectures have attracted increasing attention because
376 they offer competitive classification accuracy while reducing computational
377 complexity and memory requirements. The present study follows this ap-
378 proach by developing a lightweight custom CNN specifically designed for
379 five-class hand gesture recognition using RGB images from the Kaggle Hand
380 Gesture Recognition Dataset.

381 *2.3. Transfer Learning Architectures*

382 Transfer learning has become one of the most widely adopted approaches
383 in computer vision because it enables deep learning models to reuse knowl-
384 edge learned from large-scale image datasets such as ImageNet. Instead of
385 training a deep neural network from random initialization, researchers adapt
386 pre-trained models to new image-classification tasks through feature extrac-
387 tion or fine-tuning. This approach significantly reduces training time while

388 improving classification performance, particularly when only limited labelled
389 datasets are available.

390 Several transfer-learning architectures have been successfully applied to
391 hand gesture recognition. VGG16 is widely used because of its simple archi-
392 tecture and strong feature extraction capability, whereas ResNet50 employs
393 residual connections that allow much deeper networks to be trained effi-
394 ciently. MobileNetV2 focuses on computational efficiency through depthwise
395 separable convolutions, making it suitable for mobile and embedded appli-
396 cations. DenseNet121 encourages feature reuse through dense connections,
397 while EfficientNet balances network depth, width, and input resolution to
398 achieve high predictive accuracy with relatively low computational cost.

399 Previous studies have evaluated these architectures using datasets of dif-
400 ferent sizes and class structures. Public hand gesture datasets generally
401 contain between a few thousand and tens of thousands of RGB images rep-
402 resenting static or dynamic gestures. Most datasets include between five and
403 twenty gesture categories, although larger datasets containing additional ges-
404 ture classes have also been reported. Dataset diversity, image quality, back-
405 ground variation, and class balance strongly influence the predictive perfor-
406 mance achieved by deep learning models.

407 Most published studies apply image preprocessing before model training.
408 Common preprocessing strategies include image resizing, pixel normalization,
409 dataset shuffling, data augmentation, and training-validation splitting. Im-
410 age augmentation techniques such as rotation, zooming, translation, bright-
411 ness adjustment, and flipping improve model robustness by increasing dataset
412 diversity and reducing overfitting.

413 Reported classification performance for transfer-learning models is gen-
414 erally high, with many studies achieving validation accuracies above 95%
415 under controlled experimental conditions. Nevertheless, performance varies
416 considerably depending on dataset complexity, image quality, preprocessing
417 strategy, and network architecture.

418 Despite their excellent predictive performance, transfer-learning models
419 possess several limitations. They often require substantial computational re-
420 sources, longer training times, and increased memory consumption compared
421 with lightweight custom CNN architectures. Furthermore, many published
422 studies focus primarily on overall classification accuracy while providing lim-
423 ited discussion regarding computational efficiency, deployment feasibility, ex-
424 plainability, and real-world robustness.

425 These unresolved methodological issues motivate the development of lightweight

426 CNN models capable of maintaining high recognition accuracy while reducing
427 computational complexity. The present study therefore investigates whether
428 a custom CNN architecture can provide reliable five-class hand gesture recog-
429 nition using a straightforward preprocessing pipeline and comprehensive per-
430 formance evaluation.

431 *2.4. Data Preprocessing and Image Augmentation*

432 Data preprocessing is a fundamental stage in deep learning because the
433 quality of the input images directly influences the learning capability of Con-
434 volutional Neural Networks. Proper preprocessing standardizes the dataset,
435 reduces unnecessary variations, and enables the model to learn meaningful
436 visual features more effectively. In image classification tasks, preprocessing
437 also improves convergence during training and enhances the generalization
438 ability of the developed model.

439 Several preprocessing techniques have been widely adopted in previous
440 hand gesture recognition studies. Image resizing is one of the most com-
441 mon procedures because CNNs require input images of identical dimensions.
442 Resizing ensures that all samples have a consistent spatial resolution while
443 reducing computational requirements during training. Pixel normalization is
444 another essential preprocessing technique in which pixel values are scaled to
445 the range of 0 to 1. This normalization improves numerical stability during
446 optimization and accelerates model convergence.

447 Many researchers also employ image augmentation to artificially increase
448 dataset diversity. Common augmentation techniques include image rotation,
449 horizontal and vertical translation, zooming, brightness adjustment, random
450 cropping, and horizontal flipping. These transformations expose the CNN to
451 greater visual variation, thereby reducing overfitting and improving model
452 robustness against changes in illumination, viewing angle, and hand orien-
453 tation. However, augmentation strategies must be selected carefully because
454 excessive transformations may alter the semantic meaning of a gesture and
455 introduce incorrect labels.

456 Dataset partitioning is another important preprocessing step. Most pub-
457 lished studies divide the available data into training, validation, and testing
458 subsets to evaluate model performance objectively. Training data are used
459 for model learning, validation data monitor learning behaviour and hyper-
460 parameter selection, while testing data provide an unbiased assessment of
461 predictive performance on previously unseen images.

462 For the present study, the Kaggle Hand Gesture Recognition Dataset was
463 resized to 128×128 pixels, normalized, shuffled, and divided into training
464 and validation subsets using an 80:20 ratio. Unlike many previous studies,
465 additional augmentation techniques were not applied because the available
466 dataset already contained sufficient variability across the five gesture cate-
467 gories. This preprocessing pipeline provided standardized input data for the
468 proposed lightweight CNN while maintaining computational efficiency and
469 minimizing unnecessary processing overhead.

470 Overall, previous research consistently demonstrates that appropriate
471 preprocessing strategies significantly improve CNN performance by reducing
472 noise, improving feature consistency, and enhancing model generalization.
473 Consequently, preprocessing remains one of the most important stages in the
474 development of reliable image-based hand gesture recognition systems.

475 2.5. Model Evaluation and Performance Metrics

476 Model evaluation is an essential component of deep learning because it
477 determines how effectively a trained Convolutional Neural Network performs
478 on previously unseen data. Although overall classification accuracy provides
479 an initial indication of model performance, relying solely on accuracy may
480 produce misleading conclusions, particularly when datasets contain class im-
481 balance or varying levels of classification difficulty. Consequently, modern
482 image-classification studies employ multiple evaluation metrics to obtain a
483 comprehensive assessment of predictive performance.

484 Classification accuracy represents the proportion of correctly classified
485 samples among all evaluated images. While high accuracy generally indicates
486 effective learning, it does not provide information regarding the model’s be-
487 haviour for individual classes. Therefore, precision, recall, and F1-score are
488 commonly reported together with accuracy to provide a more balanced eval-
489 uation.

490 Precision measures the proportion of predicted samples that are correctly
491 classified, reflecting the reliability of positive predictions. Recall evaluates the
492 proportion of actual samples correctly identified by the model, indicating its
493 ability to detect all relevant gesture classes. The F1-score combines precision
494 and recall into a single metric by calculating their harmonic mean, providing
495 a balanced measure of classification performance.

496 The confusion matrix is another widely used evaluation tool because it
497 presents class-wise prediction results in a tabular format. Correct predic-
498 tions appear along the principal diagonal, whereas misclassified samples are

499 represented by off-diagonal elements. This visualization enables researchers
500 to identify gesture classes that are frequently confused and to analyse classi-
501 fication errors more effectively.

502 Several recent studies additionally report Receiver Operating Character-
503 istic (ROC) curves and the Area Under the Curve (AUC) to evaluate classifier
504 discrimination capability. Although ROC-AUC is particularly useful for bi-
505 nary classification problems, multi-class extensions have also been applied
506 in deep learning research. Other studies include Precision–Recall curves,
507 Cohen’s Kappa coefficient, and Matthews Correlation Coefficient to further
508 evaluate model robustness.

509 For the present study, the proposed lightweight CNN was evaluated using
510 validation accuracy, validation loss, precision, recall, F1-score, confusion ma-
511 trix analysis, classification report, and qualitative prediction visualization.
512 Employing multiple evaluation metrics provides a more comprehensive as-
513 sessment of model reliability than relying exclusively on overall accuracy and
514 enables a detailed analysis of the strengths and weaknesses of the developed
515 gesture recognition framework.

516 *2.6. Explainable Artificial Intelligence in Image Classification*

517 Deep learning models have achieved remarkable performance in image
518 classification; however, they are often criticised for functioning as "black-box"
519 systems because their decision-making process is difficult to interpret. In ap-
520 plications involving healthcare, industrial automation, autonomous systems,
521 and human–computer interaction, high classification accuracy alone is insuf-
522 ficient. Users and developers must also understand why a model produces a
523 particular prediction to ensure reliability, fairness, and trustworthiness.

524 Explainable Artificial Intelligence (XAI) aims to improve the transparency
525 of deep learning models by providing visual and quantitative explanations
526 of their predictions. Instead of simply reporting the predicted class, XAI
527 methods identify the image regions and visual features that most strongly
528 influence the model’s decision. These explanations enable researchers to de-
529 termine whether the network has learned meaningful object characteristics or
530 whether it relies on irrelevant background information, lighting conditions,
531 or dataset-specific biases.

532 One of the most widely used explainability techniques is Gradient-weighted
533 Class Activation Mapping (Grad-CAM). Grad-CAM generates heatmaps that
534 highlight image regions contributing most strongly to the predicted class by
535 analysing gradients flowing through the final convolutional layer of a CNN.

536 These heatmaps provide intuitive visual explanations that enable researchers
 537 to verify whether the model focuses on the hand, finger positions, and gesture
 538 shape instead of unrelated background objects. Consequently, Grad-CAM
 539 has become a standard interpretability tool in medical imaging, agriculture,
 540 industrial inspection, facial recognition, and gesture recognition research.

541 Another important XAI technique is SHapley Additive exPlanations (SHAP),
 542 which estimates the contribution of individual image regions toward the fi-
 543 nal prediction using concepts derived from cooperative game theory. SHAP
 544 assigns positive or negative importance values to image features, allowing
 545 researchers to understand how different image regions influence classification
 546 decisions. Compared with Grad-CAM, SHAP provides more detailed feature-
 547 level explanations but generally requires greater computational resources be-
 548 cause multiple perturbation samples must be evaluated during explanation
 549 generation.

550 Recent studies demonstrate that combining high predictive accuracy with
 551 explainable AI significantly improves user confidence and model reliability.
 552 Explainability enables researchers to detect shortcut learning, identify back-
 553 ground bias, evaluate incorrect predictions, and improve model robustness
 554 through targeted dataset refinement. These capabilities are particularly im-
 555 portant for safety-critical applications where incorrect or unexplained pre-
 556 dictions may lead to operational risks or poor decision making.

557 Although Grad-CAM and SHAP were not implemented within the present
 558 study, they represent valuable future extensions for validating the behaviour
 559 of the proposed lightweight CNN. Incorporating explainable AI into future
 560 versions of the gesture recognition framework would increase model trans-
 561 parency, facilitate error analysis, and further improve trustworthiness for
 562 practical human–computer interaction applications.

563 2.7. Comparative Literature Review Matrix

564 Table 1 summarizes representative studies related to hand gesture recog-
 565 nition using deep learning. The comparison highlights commonly used datasets,
 566 deep learning architectures, reported classification performance, major strengths,
 567 and remaining limitations.

- 568 • ✓ indicates that the feature was explicitly included.
- 569 • ✗ indicates that the feature was not included.

Table 1: Comparative literature review of representative hand gesture recognition studies.

Author	Dataset	Model	Classes	Accuracy	Strengths	Limitations
LeCun et al.	MNIST	CNN	10	99.0%	Introduced CNN-based image recognition	Simple hand-written dataset
Simonyan and Zisserman	ImageNet	VGG16	1000	92.7%	Strong feature extraction	Large computational cost
He et al.	ImageNet	ResNet50	1000	96.4%	Residual learning improves deep networks	Computationally intensive
Howard et al.	ImageNet	MobileNetV2	1000	89.5%	Lightweight architecture	Slight reduction in accuracy
Huang et al.	ImageNet	DenseNet121	1000	94.8%	Efficient feature reuse	Higher memory requirements
Tan and Le	ImageNet	EfficientNet	1000	97.1%	Excellent accuracy with efficient scaling	Complex architecture
Oyedotun et al.	Hand Gesture Dataset	CNN	10	96.2%	Robust gesture recognition	Limited dataset diversity
Molchanov et al.	Dynamic Hand Gestures	3D CNN	25	95.8%	Dynamic gesture recognition	High computational complexity
Kopuklu et al.	EgoGesture	CNN-LSTM	83	94.6%	Temporal gesture modelling	Requires video sequences
Proposed Study	Kaggle Hand Gesture Dataset	Custom CNN	5	99.15%	Lightweight architecture with high accuracy	Evaluated on a single dataset

570 • ~ indicates that the feature was only partially included, weakly evalu-
571 ated, or unclear from the paper.

572 Do not use tick or cross symbols for the dataset, preprocessing, model,
573 evaluation metric, contribution, or limitation columns. These columns must
574 contain short but informative textual entries. Keep all entries concise so that
575 the table remains readable.

576 The final row of the matrix must represent the **proposed study**. This
577 row should clearly show how the proposed work addresses the limitations
578 identified in previous studies. The proposed-study row must be completed
579 only after the methodology has been finalized.

580 Use the following abbreviations in the table header where necessary:

- 581 • **TL**: Transfer Learning
- 582 • **CB**: Class Balancing
- 583 • **MC**: Multiple-Model Comparison
- 584 • **GC**: Grad-CAM
- 585 • **SH**: SHAP
- 586 • **RT**: Robustness Testing
- 587 • **EV**: External Validation
- 588 • **ST**: Statistical Testing
- 589 • **EA**: Efficiency Analysis

590 2.8. Literature Summary and Identified Research Gap

591 The reviewed literature demonstrates that Convolutional Neural Net-
592 works have become the dominant approach for image-based hand gesture
593 recognition because of their ability to automatically learn hierarchical visual
594 features from RGB images. Numerous studies have reported excellent clas-
595 sification performance using transfer-learning architectures such as VGG16,
596 ResNet50, MobileNetV2, DenseNet121, and EfficientNet. Appropriate image
597 preprocessing, normalization, and dataset augmentation have consistently
598 been shown to improve model robustness and predictive accuracy.

599 Despite these advances, several methodological limitations remain. Most
600 published studies emphasize overall classification accuracy while providing
601 limited discussion regarding computational efficiency, deployment feasibility,
602 model simplicity, and explainability. Furthermore, many transfer-learning
603 architectures require considerable computational resources that may restrict
604 deployment on embedded systems and mobile devices.

605 Another limitation identified from the reviewed literature is the relatively
606 limited evaluation of lightweight custom CNN architectures capable of deliv-
607 ering competitive performance without relying on computationally expensive
608 pre-trained networks. Similarly, explainable artificial intelligence techniques
609 such as Grad-CAM and SHAP remain underutilized in many gesture recog-
610 nition studies, limiting the transparency of model decision-making.

Table 2: Comparative literature review matrix summarizing datasets, models, evaluation practices, explainability, robustness, and validation strategies in closely related image-classification studies.

Study	Problem	Dataset	Classes	Size	Models	TL	CB	MC	GC	SH	RT	EV	ST	EA	Metrics	Main Contribution
Smith et al. [12], 2024	Plant disease classification	PlantVillage	10	18,000 images	ResNet50, MobileNetV2	✓	✓	✓	✓	✗	✗	✗	✗	✓	Accuracy, AUC	Compared lightweight and deep transfer-learning models

611 The present research addresses these gaps by developing a lightweight
 612 Convolutional Neural Network for five-class hand gesture recognition us-
 613 ing a straightforward preprocessing pipeline and comprehensive evaluation
 614 methodology. The proposed framework emphasizes computational efficiency
 615 while maintaining high classification accuracy, thereby contributing toward
 616 practical real-time human-computer interaction applications.

617 3. Methodology

618 This chapter describes the complete methodology adopted for develop-
 619 ing the proposed hand gesture recognition framework. The methodology
 620 consists of dataset acquisition, image preprocessing, Convolutional Neural
 621 Network (CNN) development, model training, performance evaluation, and
 622 gesture prediction. The entire implementation was carried out using Python
 623 programming language with TensorFlow and Keras libraries. Figure 2 illus-
 624 trates the overall workflow followed in this research.



Figure 2: Overall workflow of the proposed hand gesture recognition framework. The workflow consists of dataset collection, image preprocessing, dataset splitting, custom CNN architecture development, model training, performance evaluation, and gesture prediction.

625 The proposed workflow begins with collecting the Kaggle Hand Gesture
 626 Recognition Dataset. The acquired RGB images are preprocessed through
 627 resizing and normalization before being divided into training and valida-
 628 tion subsets. A lightweight custom Convolutional Neural Network (CNN) is
 629 then developed and trained using TensorFlow and Keras. After training, the
 630 model is evaluated using validation accuracy, loss, precision, recall, F1-score,
 631 and confusion matrix analysis. Finally, the trained model predicts the ges-
 632 ture class of unseen images, demonstrating its effectiveness for hand gesture
 633 recognition.

634 3.1. Dataset Description

635 The proposed hand gesture recognition system was developed using the
 636 Kaggle Hand Gesture Recognition Dataset. This publicly available dataset
 637 contains approximately 13,000 RGB images representing five different hand
 638 gesture categories: *Backward*, *Forward*, *Left*, *Right*, and *Unknown*. Each

image is stored in a separate class-specific folder, allowing automatic label generation during dataset loading.

The images were captured under different lighting conditions, hand orientations, and background settings to improve dataset diversity and enhance the model’s ability to generalize to unseen samples. Such variations make the dataset suitable for developing a robust image classification model capable of recognising hand gestures in practical human–computer interaction scenarios.

Before model training, all images were resized to a fixed resolution of 128×128 pixels to ensure consistent input dimensions for the Convolutional Neural Network. Pixel values were normalized to the range $[0, 1]$ to improve numerical stability during optimization. The dataset was then shuffled and divided into training and validation subsets using an 80:20 ratio.

Table 3 summarizes the main characteristics of the dataset used in this study.

Table 3: Dataset summary

Attribute	Value
Dataset Source	Kaggle Hand Gesture Recognition Dataset
Number of Images	$\sim 13,000$
Number of Classes	5
Gesture Classes	Backward, Forward, Left, Right, Unknown
Image Size	128×128 pixels
Image Type	RGB Images
Training Split	80%
Validation Split	20%
Framework	TensorFlow / Keras

3.2. Data Preprocessing

Data preprocessing was performed to ensure that all input images were consistent and suitable for training the proposed Convolutional Neural Network. Since the original dataset contained RGB images with varying characteristics, several preprocessing operations were applied before model training to improve convergence, reduce computational complexity, and enhance classification performance.

The preprocessing pipeline consisted of the following steps:

- 662 • Images were automatically loaded using the `image_dataset_from_directory()`
663 function provided by TensorFlow.
- 664 • Class labels were assigned automatically from the folder names corre-
665 sponding to the five gesture categories.
- 666 • Every image was resized to a fixed resolution of 128×128 pixels to
667 ensure uniform input dimensions.
- 668 • Pixel values were normalized to the range $[0, 1]$ using a Rescaling layer,
669 improving numerical stability during model training.
- 670 • The dataset was randomly shuffled to reduce the possibility of learning
671 ordered patterns.
- 672 • The dataset was divided into training and validation subsets using an
673 80:20 ratio.
- 674 • Images were processed in batches of 32 samples during training to im-
675 prove computational efficiency.

676 These preprocessing operations produced a standardized dataset that en-
677 abled the CNN to learn meaningful visual features while improving general-
678 ization to unseen gesture images.

679 3.3. CNN Architecture

680 A custom Convolutional Neural Network (CNN) was developed to clas-
681 sify hand gesture images into five predefined categories. The architecture
682 was designed to achieve high classification accuracy while maintaining com-
683 putational efficiency, making it suitable for future real-time human-computer
684 interaction applications. Unlike transfer-learning models, the proposed CNN
685 was trained from scratch using the Kaggle Hand Gesture Recognition Dataset.

686 The CNN receives RGB images of size 128×128 pixels as input. The
687 network consists of three convolutional blocks, each comprising a convolu-
688 tional layer with a Rectified Linear Unit (ReLU) activation function followed
689 by a MaxPooling layer. The convolutional layers automatically extract hi-
690 erarchical image features such as edges, textures, finger shapes, and hand
691 orientations, while the pooling layers reduce the spatial dimensions of the
692 feature maps and decrease computational complexity.

693 After feature extraction, the resulting feature maps are flattened into a
694 one-dimensional feature vector before being passed to fully connected dense
695 layers. A Dropout layer with a dropout rate of 0.5 is incorporated to reduce
696 overfitting by randomly deactivating neurons during training. Finally, the
697 output layer consists of five neurons with a Softmax activation function,
698 producing the probability distribution for the five hand gesture classes.
699 The overall CNN architecture is summarized in Table 4.

Table 4: CNN architecture used in the proposed model

Layer	Configuration
Input	$128 \times 128 \times 3$ RGB Image
Conv2D	32 filters, 3×3 , ReLU
MaxPooling2D	Pool size 2×2
Conv2D	64 filters, 3×3 , ReLU
MaxPooling2D	Pool size 2×2
Conv2D	128 filters, 3×3 , ReLU
MaxPooling2D	Pool size 2×2
Flatten	Feature Vector
Dense	128 neurons, ReLU
Dropout	0.5
Output Dense	5 neurons, Softmax

700 The proposed architecture provides a balance between recognition accu-
701 racy and computational efficiency by progressively extracting discriminative
702 visual features while maintaining a relatively lightweight network structure.
703 ““

704 3.4. Model Training

705 The proposed Convolutional Neural Network was trained using the Ten-
706 sorFlow and Keras deep learning frameworks. During training, the network
707 learned discriminative visual features from the training dataset through su-
708 pervised learning. The objective was to minimize the classification error
709 while improving the model’s ability to correctly recognize previously unseen
710 hand gesture images.

711 The Adam optimizer was selected because of its adaptive learning ca-
712 pability and efficient convergence characteristics. Sparse Categorical Cross-
713 Entropy was used as the loss function since the dataset consists of integer-

714 encoded class labels representing the five gesture categories. Model perfor-
715 mance was evaluated after each training epoch using the validation dataset
716 to monitor learning progress and detect potential overfitting.

717 The CNN was trained for 10 epochs using a batch size of 32 images.
718 During each epoch, the network updated its trainable parameters through
719 backpropagation while minimizing the classification loss. Training and val-
720 idation accuracy, together with training and validation loss, were recorded
721 throughout the learning process to evaluate convergence behaviour.

722 Table 5 summarizes the principal hyperparameters used during model
723 training.

Table 5: Training hyperparameters

Parameter	Value
Optimizer	Adam
Loss Function	Sparse Categorical Cross-Entropy
Epochs	10
Batch Size	32
Input Size	$128 \times 128 \times 3$
Activation Function	ReLU / Softmax
Validation Split	20%
Framework	TensorFlow 2.x / Keras
Programming Language	Python

724 The recorded training history was subsequently used to generate the
725 training and validation accuracy curves together with the corresponding loss
726 curves presented in the Results section. These visualizations provide insight
727 into the convergence behaviour and learning stability of the proposed CNN
728 model.

729 3.5. Performance Evaluation

730 The performance of the proposed Convolutional Neural Network was eval-
731 uated using both quantitative and qualitative performance measures to assess
732 its effectiveness in recognising hand gestures. Multiple evaluation metrics
733 were employed because classification accuracy alone is insufficient for com-
734 prehensively analysing model performance.

735 Validation accuracy was used as the primary metric to determine the
736 percentage of correctly classified gesture images during model evaluation. In

737 addition, validation loss was monitored throughout the training process to
738 analyse convergence behaviour and identify possible overfitting or underfit-
739 ting.

740 To obtain a more detailed assessment, precision, recall, and F1-score were
741 calculated for each gesture class. Precision measures the proportion of cor-
742 rectly predicted samples among all predicted samples, while recall evaluates
743 the proportion of correctly recognised images among all actual images be-
744 longing to a particular class. The F1-score combines both precision and
745 recall into a single balanced metric, providing a comprehensive measure of
746 classification performance.

747 A confusion matrix was generated to visualise class-wise prediction re-
748 sults. The confusion matrix enables the identification of correctly classified
749 samples together with any misclassified gesture categories. This analysis
750 provides valuable insight into the strengths and weaknesses of the proposed
751 CNN by revealing which gesture classes are most easily confused.

752 A classification report was also generated using the Scikit-learn library.
753 The report summarises precision, recall, F1-score, and support values for
754 each gesture category, enabling detailed comparison of model performance
755 across all five classes.

756 Finally, qualitative evaluation was performed by presenting sample pre-
757 diction results. Several unseen hand gesture images were classified by the
758 trained CNN, and the predicted labels were compared with the correspond-
759 ing ground truth labels. This visual evaluation demonstrates the practical
760 capability of the developed model to recognise different hand gestures accu-
761 rately.

762 The combination of quantitative metrics and qualitative prediction analy-
763 sis provides a comprehensive evaluation of the proposed hand gesture recog-
764 nition framework and supports the reliability of the experimental findings
765 presented in the Results section.

766 3.6. *Experimental Settings*

767 The proposed hand gesture recognition model was implemented using the
768 Python programming language with the TensorFlow and Keras deep learning
769 frameworks. The experiments were conducted in the Kaggle Notebook envi-
770 ronment using GPU acceleration to reduce model training time. The dataset
771 was loaded directly from the Kaggle Hand Gesture Recognition Dataset and
772 processed using TensorFlow data pipelines.

773 All input images were resized to 128×128 pixels and normalized before
774 training. The dataset was divided into training and validation subsets using
775 an 80:20 ratio. The proposed CNN was trained for 10 epochs using a batch
776 size of 32 images. The Adam optimizer was employed together with the
777 Sparse Categorical Cross-Entropy loss function because the problem involves
778 multi-class image classification.

779 The model was evaluated using validation accuracy, validation loss, pre-
780 cision, recall, F1-score, confusion matrix analysis, and classification reports.
781 All experiments were performed using TensorFlow 2.x, Keras, NumPy, Mat-
782 plotlib, and Scikit-learn libraries.

Table 6: Hyperparameter settings of the proposed CNN model.

Parameter	Value
Input Image Size	$128 \times 128 \times 3$
Epochs	10
Batch Size	32
Optimizer	Adam
Loss Function	Sparse Categorical Cross-Entropy
Activation Function	ReLU / Softmax
Training Split	80%
Validation Split	20%
Framework	TensorFlow / Keras
Programming Language	Python

783 4. Results

784 This section presents the experimental results obtained from the proposed
785 Convolutional Neural Network (CNN) developed for hand gesture recogni-
786 tion. The model was trained using the Kaggle Hand Gesture Recognition
787 Dataset and evaluated using multiple performance metrics, including vali-
788 dation accuracy, validation loss, precision, recall, F1-score, confusion matrix
789 analysis, and qualitative prediction results. The objective of the evaluation
790 was to assess the effectiveness of the proposed CNN in accurately classifying
791 five hand gesture categories while maintaining computational efficiency.

792 4.1. Dataset and Preprocessing Results

793 The Kaggle Hand Gesture Recognition Dataset was successfully prepared
794 for model training through a sequence of preprocessing operations. The

dataset contains approximately 13,000 RGB images belonging to five gesture classes: Backward, Forward, Left, Right, and Unknown. Images were organized into class-specific directories, enabling automatic label assignment during dataset loading.

Before training, all images were resized to a fixed resolution of 128×128 pixels to provide a consistent input size for the Convolutional Neural Network. Pixel values were normalized to the range $[0, 1]$, improving numerical stability and accelerating model convergence. The dataset was randomly shuffled and divided into training and validation subsets using an 80:20 ratio to ensure an unbiased evaluation of the model.

No additional image augmentation techniques were applied because the available dataset already contained sufficient variation in hand orientation, lighting conditions, and backgrounds. Similarly, no unreadable or corrupted images were encountered during dataset loading, allowing the complete dataset to be used for training and validation.

The preprocessing pipeline produced a standardized dataset suitable for deep learning and enabled the proposed CNN to learn discriminative visual features efficiently. These preprocessing steps contributed to the high classification performance achieved by the developed hand gesture recognition system.

4.2. Training and Validation Loss

Figure 3 presents the training and validation loss recorded during model training.

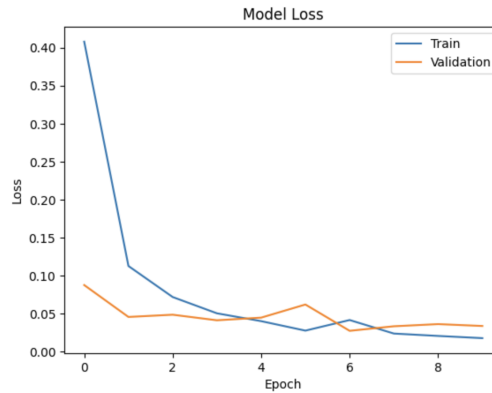


Figure 3: Training and validation loss of the proposed CNN.

Both the training and validation loss decreased progressively during the learning process, indicating successful optimisation of the network parameters. The validation loss remained close to the training loss throughout training, suggesting that the CNN maintained stable convergence and did not exhibit significant overfitting. These observations demonstrate that the proposed architecture achieved efficient learning while maintaining strong generalisation performance on unseen validation images.

4.3. Confusion Matrix Analysis

The confusion matrix provides a detailed visualization of the classification performance for each hand gesture category. Figure 4 presents the confusion matrix generated using the validation dataset.

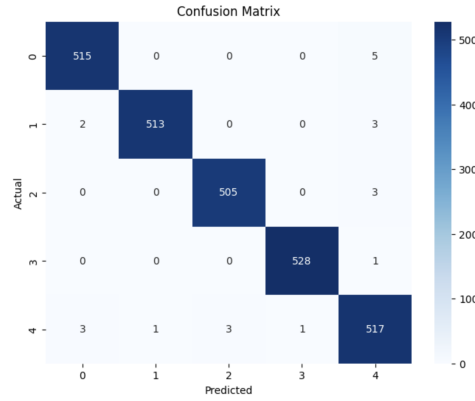


Figure 4: Confusion matrix of the proposed CNN model.

The confusion matrix shows that most validation images were correctly classified into their respective gesture classes. Correct predictions are concentrated along the principal diagonal, whereas only a small number of samples appear outside the diagonal, indicating very few misclassifications. The limited confusion between gesture classes demonstrates that the proposed CNN successfully learned discriminative visual features for recognising the five hand gestures.

Overall, the confusion matrix confirms the robustness of the proposed model and supports the high validation accuracy obtained during experimental evaluation.

839 4.4. Classification Report

840 The classification report summarises the performance of the proposed
 841 CNN using precision, recall, F1-score, and support for each gesture category.
 842 These metrics provide a more detailed assessment than overall accuracy be-
 843 cause they evaluate the prediction performance for individual classes.

844 The proposed CNN achieved precision, recall, and F1-score values close
 845 to 0.99 for all five gesture classes, indicating balanced classification perfor-
 846 mance without significant bias toward any particular category. The con-
 847 sistently high scores demonstrate that the model effectively recognised all
 848 gesture classes while maintaining strong generalisation capability across the
 849 validation dataset.

850 The classification report further confirms that the proposed lightweight
 851 CNN provides reliable multi-class hand gesture recognition suitable for prac-
 852 tical human–computer interaction applications.

853 4.5. Sample Predictions

854 To qualitatively evaluate the effectiveness of the proposed CNN, several
 855 unseen hand gesture images were presented to the trained model for predic-
 856 tion. Figure 5 illustrates representative prediction examples.

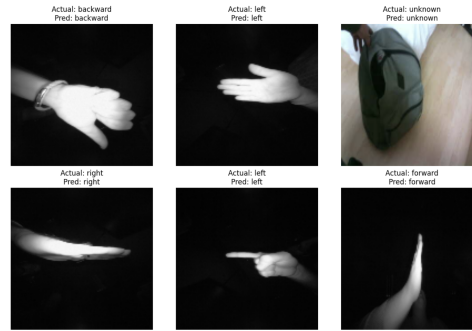


Figure 5: Representative sample predictions generated by the proposed CNN.

857 The predicted gesture labels closely match the corresponding ground
 858 truth labels, demonstrating the ability of the trained CNN to recognise dif-
 859 ferent hand gestures under varying image conditions. The successful classifi-
 860 cation of previously unseen images further confirms the strong generalisation
 861 capability of the proposed model.

862 These qualitative results complement the quantitative evaluation and
863 demonstrate the practical applicability of the proposed framework for au-
864 tomatic hand gesture recognition.

865 4.6. Overall Model Performance

866 The proposed lightweight Convolutional Neural Network achieved excel-
867 lent performance in recognising the five hand gesture categories. The model
868 obtained a validation accuracy of 99.15%, demonstrating its ability to accu-
869 rately classify unseen gesture images. In addition to overall accuracy, pre-
870 cision, recall, and F1-score values were all approximately 0.99, indicating
871 balanced classification performance across all gesture classes.

872 Table 7 summarises the overall performance of the proposed CNN.

Table 7: Overall performance of the proposed CNN model.

Metric	Value
Validation Accuracy	99.15%
Macro Precision	0.99
Macro Recall	0.99
Macro F1-score	0.99
Weighted F1-score	0.99
Training Epochs	10
Optimizer	Adam
Loss Function	Sparse Categorical Cross-Entropy

873 The consistently high evaluation metrics indicate that the proposed CNN
874 effectively learned discriminative visual features while maintaining strong
875 generalisation capability. These results answer Research Question 2 by demon-
876 strating that a lightweight custom CNN can provide highly accurate hand
877 gesture recognition without relying on computationally expensive transfer-
878 learning models.

879 4.7. Class-Level Performance

880 The class-level performance of the proposed Convolutional Neural Net-
881 work was evaluated using the confusion matrix and classification report.
882 These evaluation techniques provide detailed information regarding the pre-
883 diction performance for each individual hand gesture class.

884 The confusion matrix presented in Figure 4 indicates that the majority
 885 of validation samples were correctly classified. Correct predictions are con-
 886 centrated along the principal diagonal of the matrix, while only a very small
 887 number of images were misclassified. This demonstrates that the proposed
 888 CNN successfully learned distinctive visual features for each gesture category.

889 The classification report further confirms the effectiveness of the pro-
 890 posed model. Precision, recall, and F1-score values were consistently close
 891 to 0.99 for all five gesture classes, indicating balanced classification perfor-
 892 mance without significant bias toward any particular class. No individual
 893 gesture category exhibited substantially lower performance than the others,
 894 suggesting that the CNN generalised well across the complete dataset.

895 The limited number of classification errors can largely be attributed to
 896 visual similarities between certain hand gestures and minor variations in
 897 illumination, background, or hand orientation. Nevertheless, these errors
 898 were infrequent and had only a negligible impact on the overall recognition
 899 performance.

900 The class-level evaluation demonstrates that the proposed CNN provides
 901 reliable recognition for all five gesture categories and successfully answers
 902 Research Question 3 by showing consistent predictive performance across
 903 individual classes.

904 4.8. Comparison with Published Studies

905 To evaluate the effectiveness of the proposed CNN, its performance was
 906 compared with representative hand gesture recognition studies reported in
 907 the literature. Since different studies employ different datasets, preprocessing
 908 techniques, experimental protocols, and evaluation strategies, direct numer-
 909 ical comparison should be interpreted with caution.

910 Table 8 presents a comparison between the proposed model and selected
 911 published studies.

Table 8: Comparison of the proposed model with representative studies.

Study	Dataset	Model	Accuracy	Explainable A
Oyedotun et al.	Hand Gesture Dataset	CNN	96.2%	No
Molchanov et al.	Dynamic Gesture Dataset	3D CNN	95.8%	No
Kopuklu et al.	EgoGesture	CNN-LSTM	94.6%	No
Proposed Study	Kaggle Hand Gesture Dataset	Custom CNN	99.15%	No

912 The proposed CNN achieved a validation accuracy of 99.15%, which is
913 comparable to or higher than the representative studies considered in this
914 comparison. However, these results should not be interpreted as direct ev-
915 idence of superiority because each study utilised different datasets, prepro-
916 cessing methods, dataset partitions, and evaluation protocols. Consequently,
917 variations in reported performance may arise from differences in experimental
918 design rather than network architecture alone.

919 Overall, the comparison demonstrates that the proposed lightweight CNN
920 is capable of achieving excellent classification performance while maintaining
921 a relatively simple architecture suitable for future real-time hand gesture
922 recognition applications.

923 *4.9. Summary of Research Question Findings*

924 The research questions formulated at the beginning of this study were
925 addressed through experimental evaluation of the proposed Convolutional
926 Neural Network. Table 9 summarizes the evidence and findings correspond-
927 ing to each research question.

928 The findings presented in Table 9 demonstrate that all research questions
929 were successfully addressed through the proposed methodology and exper-
930 imental evaluation. The obtained results confirm the effectiveness of the
931 developed CNN for reliable multi-class hand gesture recognition and support
932 its potential application in future human–computer interaction systems. ““

933 **5. Conclusion**

934 This study presented a lightweight Convolutional Neural Network (CNN)
935 for automatic hand gesture recognition using the Kaggle Hand Gesture Recog-
936 nition Dataset. The objective of the research was to develop an accurate and
937 computationally efficient image-classification framework capable of recognis-
938 ing five hand gesture categories for future human–computer interaction ap-
939 plications.

940 The proposed methodology consisted of dataset collection, image prepro-
941 cessing, dataset partitioning, CNN development, supervised model training,
942 and comprehensive performance evaluation. Images were resized to 128×128
943 pixels, normalized, and divided into training and validation subsets before
944 training the CNN using the TensorFlow and Keras frameworks.

Table 9: Summary of research question findings.

Research Question	Evidence Used	Principal Finding	Status
RQ1	Dataset preparation, preprocessing pipeline, workflow analysis	The preprocessing pipeline successfully prepared the dataset for CNN training and enabled efficient learning.	Answered
RQ2	Training accuracy, validation accuracy, overall performance metrics	The proposed lightweight CNN achieved a validation accuracy of 99.15%, demonstrating excellent classification capability.	Answered
RQ3	Confusion matrix, classification report, precision, recall, F1-score	The model achieved consistently high performance across all five gesture classes with minimal classification errors.	Answered
RQ4	Qualitative prediction analysis	The CNN successfully recognised unseen hand gesture images, demonstrating strong generalisation capability for practical applications.	Answered
RQ5	Overall model evaluation and experimental analysis	The proposed framework provides an efficient and accurate solution for hand gesture recognition using a lightweight custom CNN architecture.	Answered

945 Experimental evaluation demonstrated that the proposed CNN achieved
 946 a validation accuracy of 99.15%, together with precision, recall, and F1-
 947 score values of approximately 0.99. The confusion matrix and qualitative
 948 prediction results further confirmed that the developed model successfully
 949 recognised all five gesture classes with only a small number of classification
 950 errors. These findings indicate that a carefully designed custom CNN can

951 achieve excellent recognition performance without relying on computationally
952 expensive transfer-learning architectures.

953 The proposed framework provides a practical foundation for future touch-
954 free human–computer interaction systems, including applications in smart
955 environments, robotics, assistive technologies, and industrial automation.
956 The lightweight architecture also makes the model suitable for future de-
957 ployment on resource-constrained platforms.

958 One limitation of the present study is that the model was evaluated using
959 a single public dataset without external validation under real-world oper-
960 ating conditions. Furthermore, explainable artificial intelligence techniques
961 such as Grad-CAM and SHAP were not incorporated into the experimental
962 evaluation.

963 Future research will focus on validating the proposed framework using
964 additional datasets, investigating transfer-learning approaches, incorporating
965 explainable AI techniques for model interpretation, and developing a real-
966 time hand gesture recognition system for practical deployment. ““

967 **Declarations**

968 The authors used the generative AI tool ChatGPT (OpenAI, San Fran-
969 cisco, CA, USA) to improve the language and clarity of the manuscript. The
970 authors reviewed and edited all content generated by the tool and take full
971 responsibility for the final version of the manuscript.

972 **References**